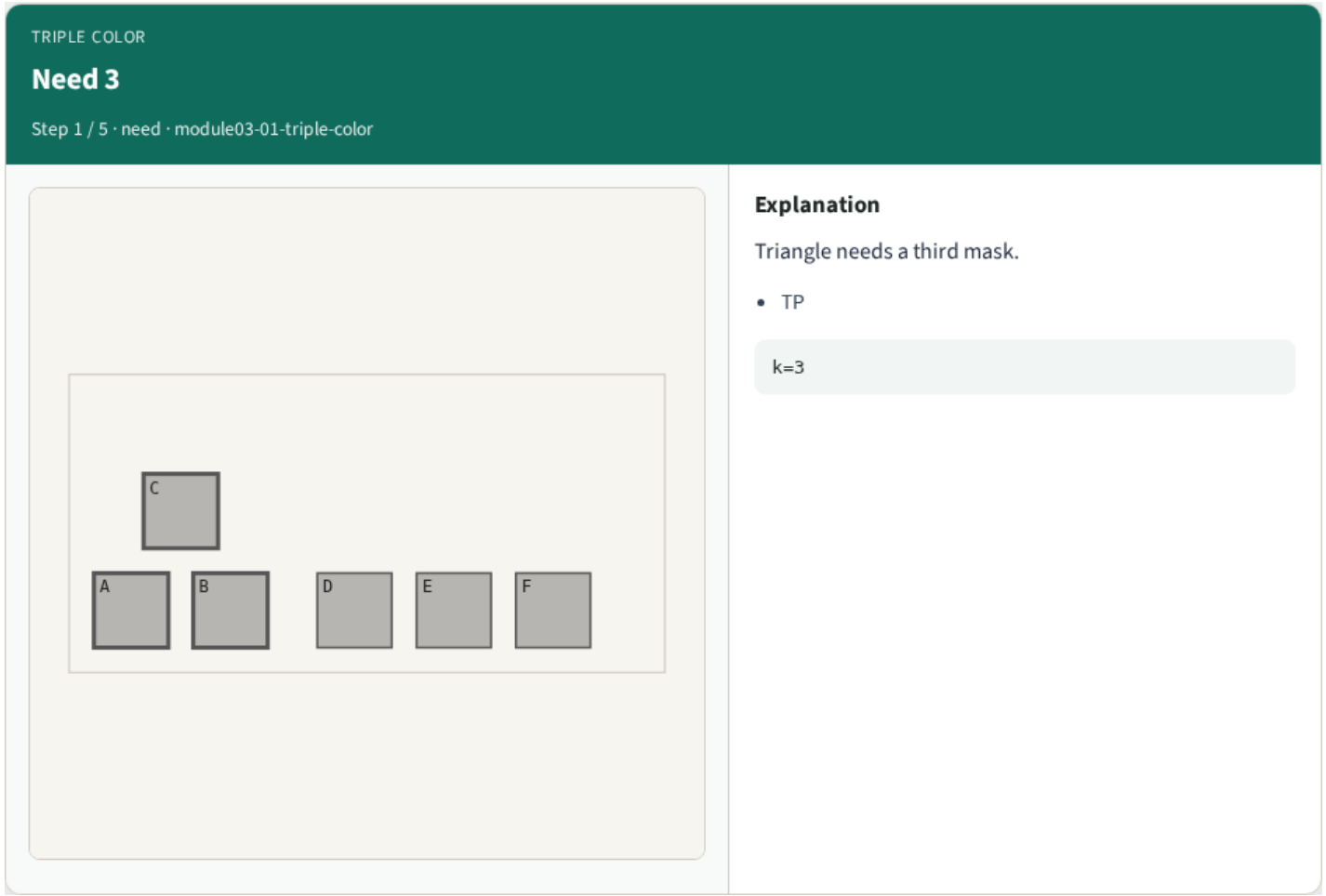
Triple patterning

When two masks fail, triple patterning assigns three masks with a 3-coloring

The idea

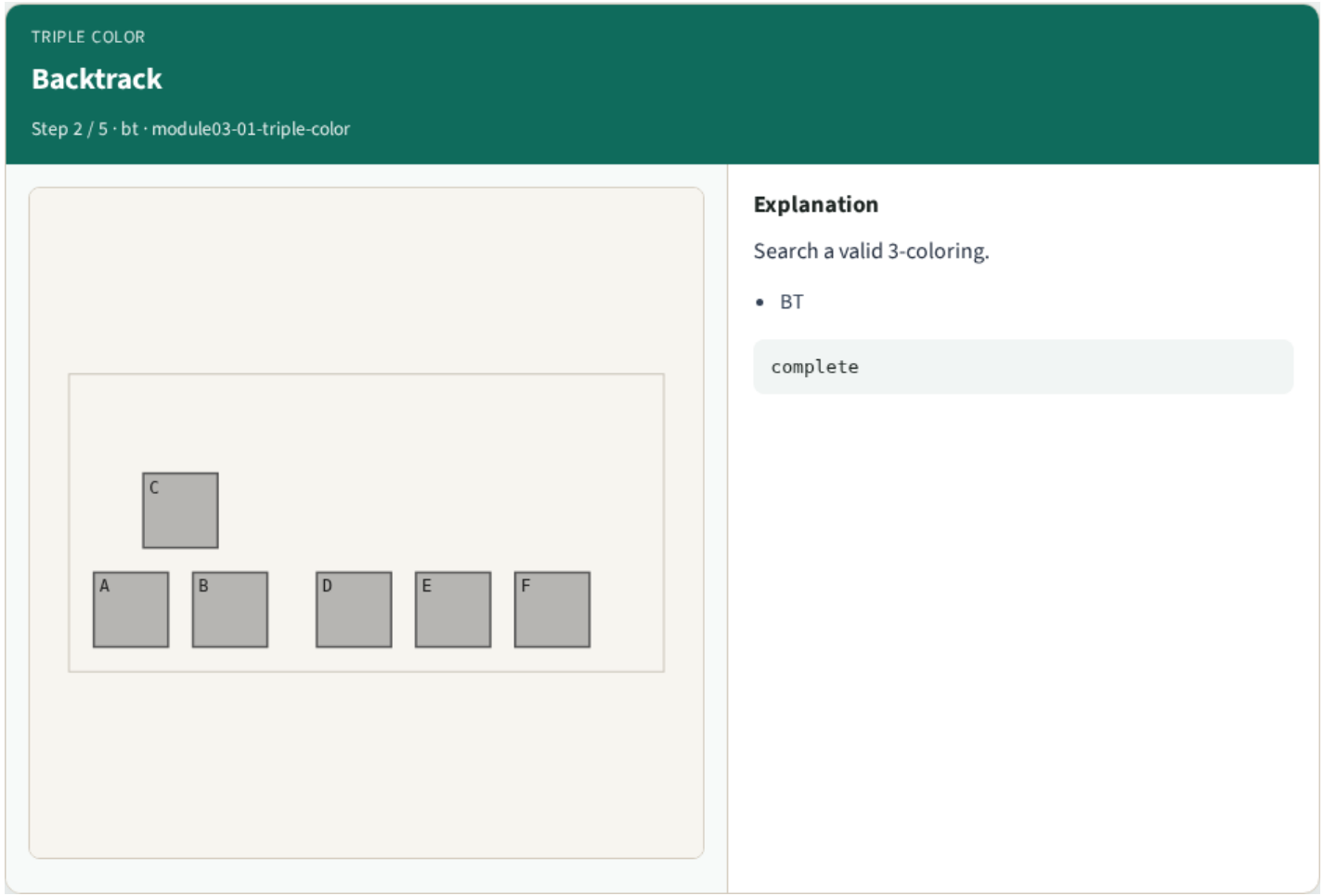
- GOLDEN: colors_used=3, ok=True on tiny_mp

Need 3


TRIPLE COLOR
Need 3
Step 1 / 5 · need · module03-01-triple-color
C
A
B
D
E
F
Explanation
Triangle needs a third mask.
TP
k=3

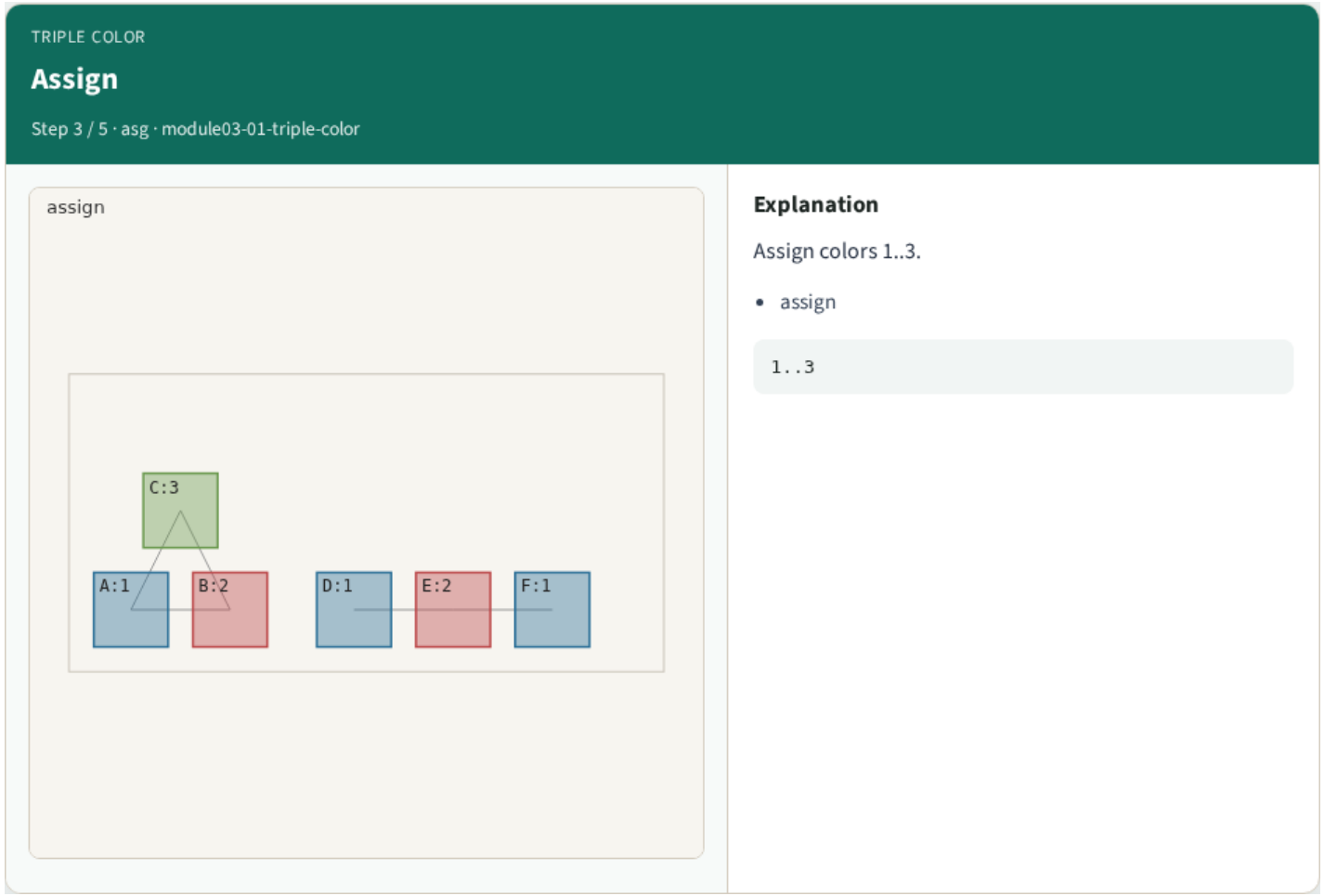
Need 3

Backtrack



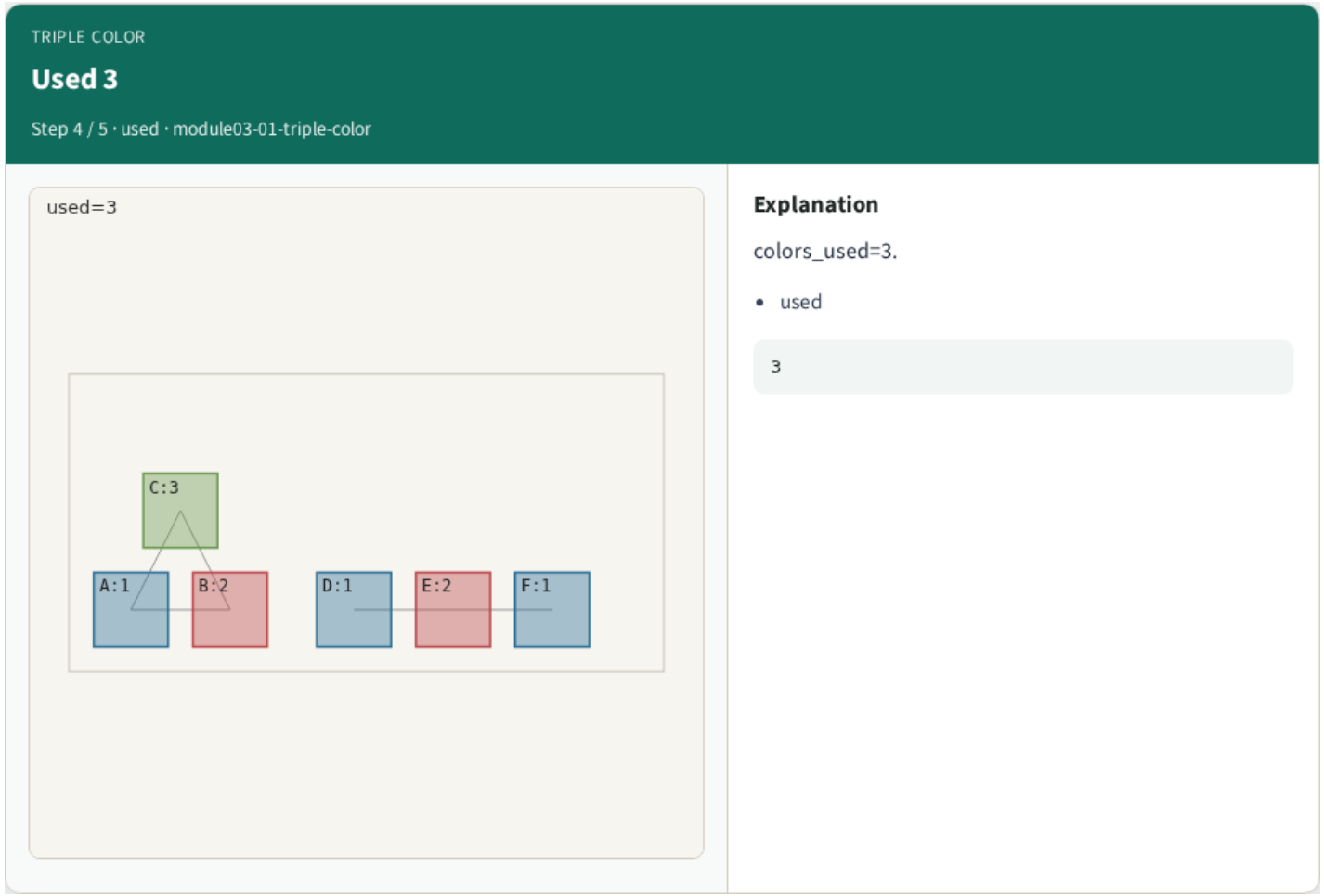
Backtrack

Assign



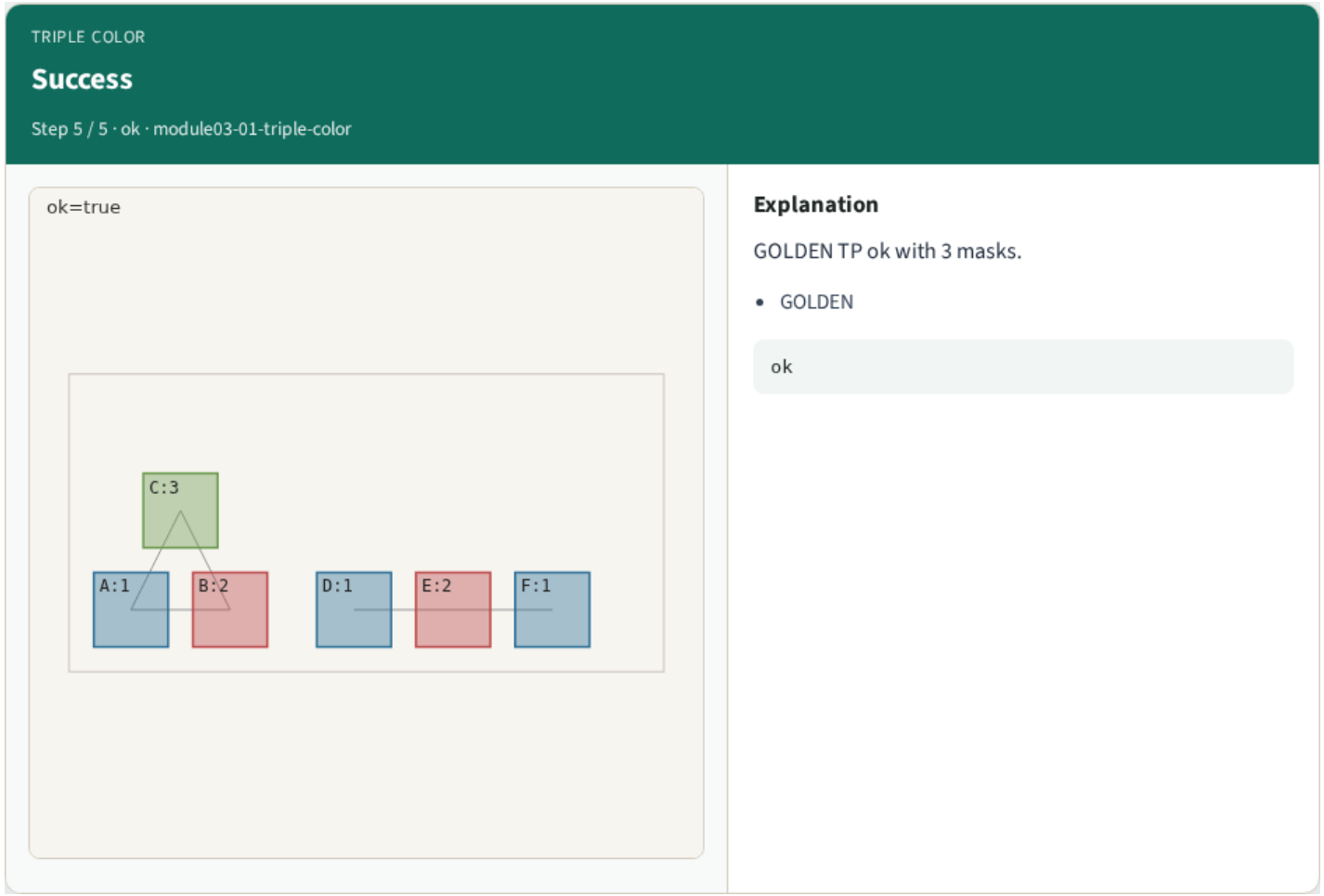
Assign

Used 3


TRIPLE COLOR
Used 3
Step 4 / 5 · used · module03-01-triple-color
used=3
C:3
A:1
B:2
D:1
E:2
F:1
Explanation
colors_used=3.
used
3

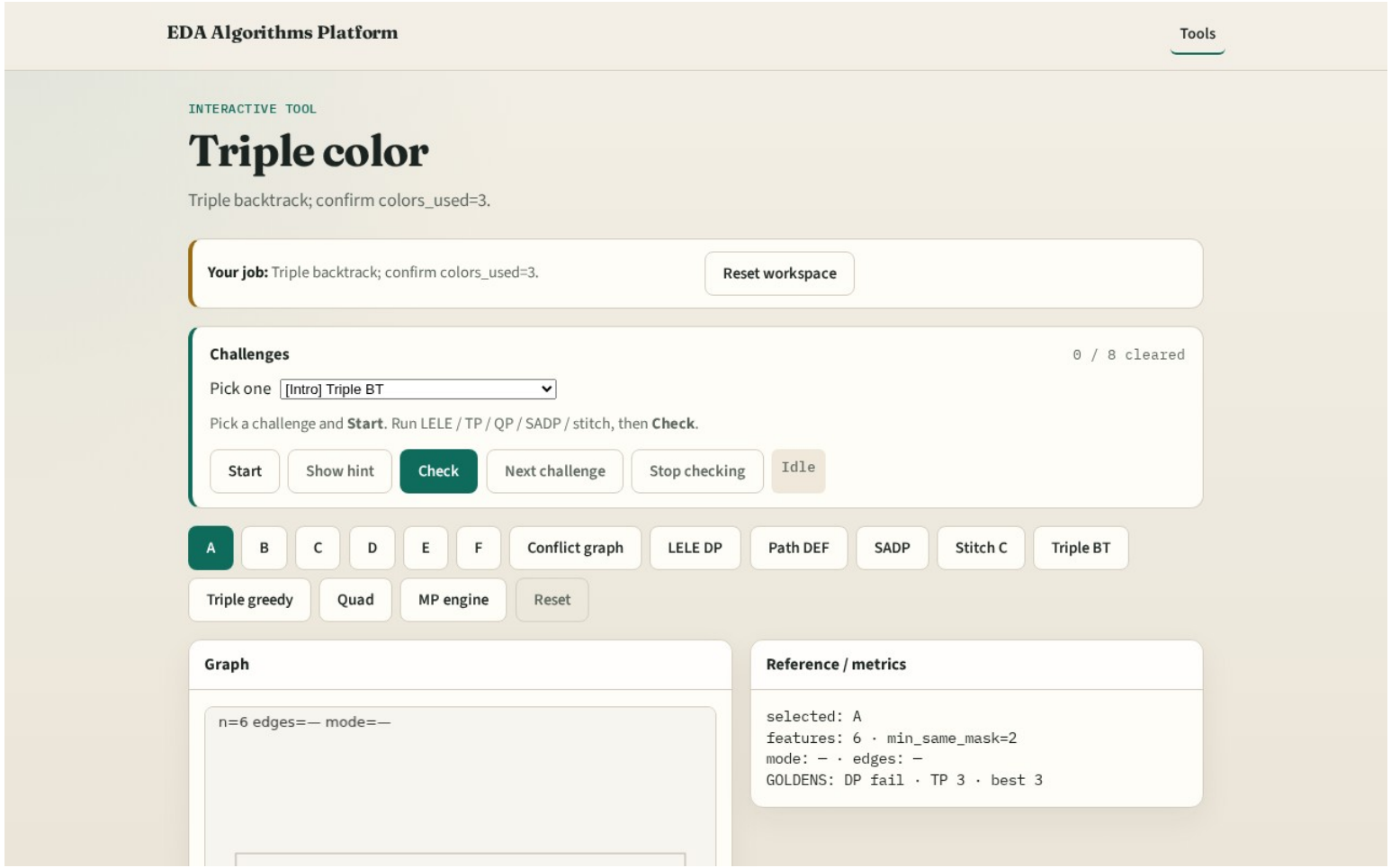
Used 3

Success



Success

Browser lab track



Browser lab starter

Implement track

- Call triple_pattern(method='backtrack'); assert ok and colors_used==3

Pitfalls

- Greedy-only when backtrack is required
- Stopping at 2 colors after stitch lesson

Your turn

- Next: compare greedy vs backtrack decompositions